RAJALAKSHMI ENGINEERING COLLEGE

Naveen Raj V 251001158-2025-IT

N2

Started on	Thursday, 24 September 2026, 10:13 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:18 AM
Time taken	5 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findDuplicate(int* nums, int numsSize) {

    int tortoise = nums[0];
    int hare = nums[0];

    do {
        tortoise = nums[tortoise];
        hare = nums[nums[hare]];
    } while (tortoise != hare);
    int ptr1 = nums[0];
    int ptr2 = tortoise;

    while (ptr1 != ptr2) {
        ptr1 = nums[ptr1];
        ptr2 = nums[ptr2];
    }

    return ptr1;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *nums = (int*)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int duplicate = findDuplicate(nums, n);
    printf("%d\n", duplicate);

    free(nums);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course